ZEN & NoiFi — Reproduction & Extension Report

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Generated: 2026-10-31 14:02:39
Repository: https://github.com/khaledhasanirfan/zen-noifi-reproduction
Hardware: MacBook Air M5, 16GB RAM, macOS 26.6.2

1. Project Overview

This report documents the reproduction and extension of two security/ML research papers:

Paper	Domain	Core Contribution
ZEN (CyFI Lab, Georgia Tech)	Memory Forensics + DNN	Recover custom neural-network layers from Linux memory dumps, attribute base models, and patch for inference
NoiFi (Shahriar et al.)	WiFi Sensing Security	Detect adversarial attacks on WiFi CSI classifiers using randomized-smoothing noise-feature analysis

2. Upstream Attribution

```
65918942e88dec93120074cc1e1963f6b07d0c35 third_party/NoiFi (heads/main)
34a0feac1ed9a93ec451d9c24cd75c12333dd0ff third_party/ZEN (heads/main)
```

3. Experiment Status Summary

#	Experiment	Status	Evidence
1	NoiFi Environment Setup	Pending	N/A
2	NoiFi Smoke Pipeline (2 epochs)	Pending	0 log files
3	NoiFi Baseline Reproduction	Pending	N/A
4	NoiFi N-vs-Stability Extension	Pending	N/A
5	ZEN Environment Setup (Linux)	Pending	N/A
6	ZEN Supplied-Dump Recovery	Pending	0 artifacts
7	ZEN Patched Inference	Pending	N/A
8	ZEN Output Consistency	Pending	N/A

4. NoiFi — Detailed Results

4.1 Environment

Component	Value
Device	CPU (Apple M5 — no CUDA)
Python	3.9.6
PyTorch	2.0.1
Dataset	RoboFiSense

4.2 Pipeline Steps

Step	Script	Purpose
0a	step_0_train_classifiers.py	Train WiFi CSI classifiers
0b	step_0_train_autoencoders.py	Train denoising autoencoders (DAE)
1	step_1_generate_adversarial_examples_simple.py	Generate adversarial attacks
2	step_2_reconstruct_and_extract_noise.py	DAE reconstruction + noise extraction
3	step_3_analyze_noise_features.py	Analyze residual noise features
4	step_4_detect_adversarial_examples_simple.py	Detect adversarial examples

4.3 Extension: Randomization Count (N) vs Detection Stability

This extension examines how the number of randomized references (N) affects detection accuracy and latency — a key design parameter in NoiFi's randomized smoothing approach.

N	AUROC	Recall@FPR	Median Latency	Score Variability
1	—	—	—	—

3	—	—	—	—
5	—	—	—	—
10	—	—	—	—
20	—	—	—	—

Table will be populated after baseline experiments complete.

5. ZEN — Detailed Results

5.1 Environment Requirements

Component	Value
OS	Linux x86-64 (required)
Python	3.8.18 (per official demo)
Framework	Volatility3 + custom CMR plugin
Model	YOLOv5 + CBAM-Involution (VisDrone)

5.2 Recovery Pipeline

Phase	Description	Status
Memory Dump	Load supplied .lime file	Pending
CMR Recovery	Extract custom model representations	Pending
Attribution	Identify base model architecture	Pending
Patching	Generate inference-ready patch	Pending
Inference	Run model with recovered patch	Pending

5.3 Extension: Output Consistency Checker

Numerically compares original model outputs against ZEN-recovered outputs using configurable absolute and relative tolerance thresholds.

```
python scripts/compare_outputs.py reference.npy recovered.npy --atol 1e-5 --rtol 1e-4
```

6. Reproduction Notes & Limitations

- NoiFi runs on CPU only (no CUDA on macOS) — training is slower but numerically equivalent
- ZEN requires Linux x86-64 with matching kernel symbols for full forensic pipeline
- All reported results are from our own runs, not copied from the papers
- Threshold tuning data and final evaluation data are kept separate
- Small epoch counts in smoke tests verify pipeline connectivity, not research claims

7. Repository Structure

```
zen-noifi-reproduction/
├── README.md
├── .gitignore
├── .gitmodules
├── third_party/
│   ├── ZEN/          (git submodule → CyFI-Lab-Public/ZEN)
│   └── NoiFi/        (git submodule → shahriar0651/NoiFi)
├── scripts/
│   ├── run_noifi_smoke.sh
│   ├── run_noifi_baseline.sh
│   ├── run_noifi_n_study.sh
│   ├── run_zen_demo.sh
│   ├── run_zen_inference.sh
│   ├── compare_outputs.py
│   └── generate_report.py
├── environments/
│   └── noifi-mac-initial.txt
├── reports/
│   ├── upstream-commits.txt
│   └── experiment_report.html
├── results/     (experiment outputs)
├── logs/        (execution logs)
└── data/        (datasets — gitignored)
```

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